

Multiple Choice (10 marks)

1. Using conditioning, how can one determine the sensitivity of a dog's hearing to sounds of different frequencies?
 - (a) By using a controlled group of dogs not exposed to the sounds
 - (b) By using sensory deprivation to heighten hearing sensitivity
 - (c) By observing the dog's natural behavior in different sound environments
 - (d) By using negative reinforcement only
 - (e) By using operant conditioning
2. How can one demonstrate that amino acids are used to build proteins in cells, while the proteins already in the cell are used in catabolism?
 - (a) Enzyme-linked immunosorbent assay (ELISA)
 - (b) Polymerase chain reaction (PCR)
 - (c) X-ray crystallography
 - (d) Mass spectrometry
 - (e) Autoradiography
3. What prevents the stomach from being digested by its own secretions?
 - (a) The thick muscular wall of the stomach
 - (b) Antibacterial properties of stomach acid
 - (c) Gastric juice
 - (d) High pH levels in the stomach
 - (e) Presence of beneficial bacteria
4. Why would you not expect conjugation among a group of paramecia that had descended from a single individual through repeated fission?
 - (a) Conjugation is prevented by the presence of a dominant allele
 - (b) Conjugation results in lesser genetic diversity
 - (c) Conjugation can occur among identical organisms
 - (d) All individuals are identical genetically, and conjugation could not occur.
 - (e) Conjugation does not require different mating types
5. If the activity of an enzyme is constant over a broad range of pH values, it is likely that

- (a) the enzyme is inactive at all pH values
- (b) no ionizing groups on the enzyme or substrate participate in the reaction
- (c) only ionizing groups on the substrate participate in the reaction
- (d) ionizing groups on both the enzyme and substrate participate in the reaction
- (e) only non-ionizing groups on the substrate participate in the reaction

True / False (10 marks)

Answer True or False

- 6. True or False: Aposomatic coloration is a strategy used by prey to camouflage themselves in their surroundings for protection.
- 7. True or False: The number of synapses crossed by a signal affects the brain's ability to detect differences in stimulus intensity.
- 8. True or False: The synthesis of an RNA/DNA hybrid from a single-stranded RNA template requires a DNA or RNA primer and reverse transcriptase.
- 9. True or False: The process of molting in crustaceans is influenced by hormonal factors rather than solely by water temperature.
- 10. True or False: Hemoglobin's affinity for O₂ increases as blood pH decreases, enhancing oxygen delivery to tissues.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the role of water availability in the transition from forest ecosystems to grasslands as one travels from east to west in the U.S., and analyze the implications for biodiversity.
- 12. Discuss the differences between kinesis and taxis in animal behavior, emphasizing how each response reflects the organism's interaction with its environment.

End of Paper